ECHOSPHERE v2.2

System Architecture, ER Diagram & Operational Methodology Manual

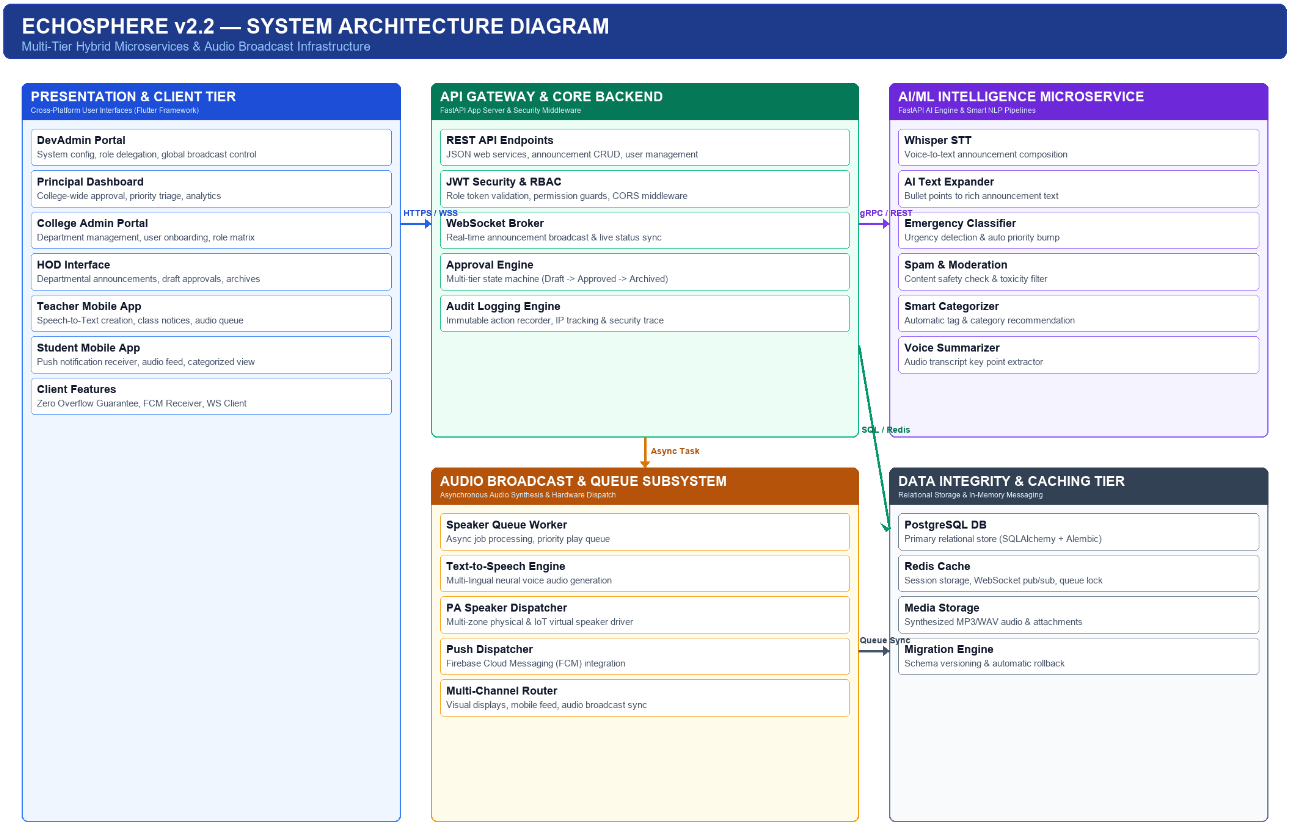
System Name: EchoSphere Campus Broadcast	**Version:** 2.2 Release
Architecture: Hybrid Microservices & Event Audio	**Primary Target:** Android Mobile APK
Database: PostgreSQL 15 + Redis Pub/Sub	**AI Subsystem:** Whisper STT & FastNLP
Document Scope: Architectural Specification, ERD, Methodology	**Classification:** Proprietary Technical Manual

EXECUTIVE SUMMARY & SYSTEM SCOPE

EchoSphere v2.2 represents a state-of-the-art multi-channel campus announcement and public address system. Engineered with a high-concurrency FastAPI core backend, a dedicated AI/ML microservice, and a cross-platform Flutter client, EchoSphere automates the entire lifecycle of campus communication—from voice-assisted drafting, spam/toxicity filtering, and multi-level role authorization to real-time physical speaker output and mobile push dispatch.

1. System Architecture Specification

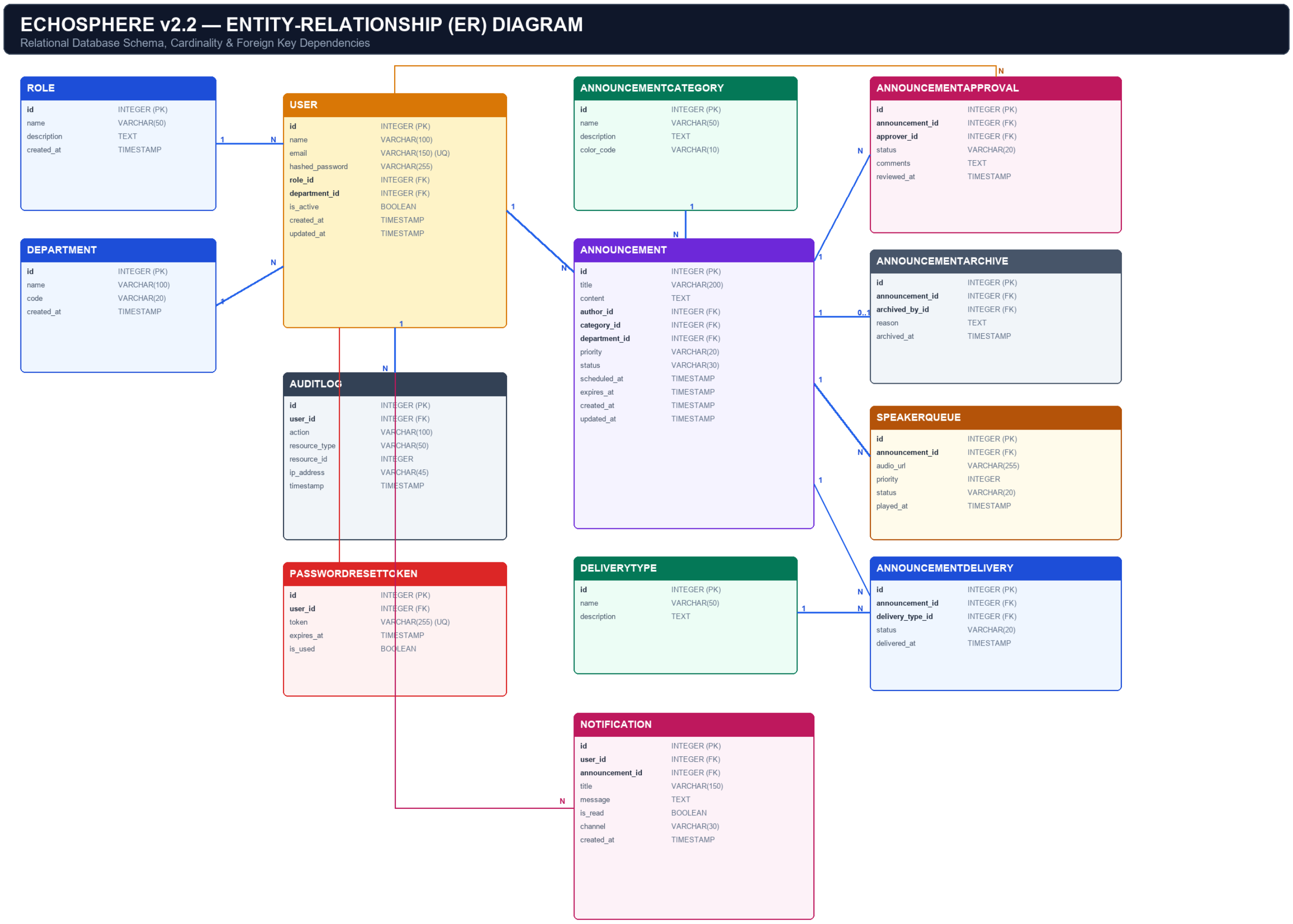
EchoSphere v2.2 employs a high-performance **Multi-Tier Hybrid Microservices Architecture** designed for scalability, zero visual overflow across mobile aspect ratios, and reliable real-time broadcast execution.



Architectural Tier	Technologies	Core Responsibilities & Components
Presentation / Client	Flutter 3.x, Dart, FCM SDK	Cross-platform mobile & web client covering 6 roles. Enforces Zero Overflow Guarantee via responsive flexible constraints.
API Gateway & Core Backend	FastAPI, Python 3.11, OAuth2/JWT, WebSockets	RESTful endpoints, JWT security middleware, RBAC enforcement, WebSocket live feed, approval flow state machine.
AI/ML Intelligence Service	FastAPI, Whisper STT, PyTorch, Transformers	Speech-to-text drafting, AI bullet text expander, emergency urgency classifier, toxicity/spam moderation, smart tag recommendation.
Audio & Broadcast Subsystem	Asyncio Workers, Neural TTS, Celery/Redis	Priority speaker queue worker, multi-lingual TTS audio synthesis, physical PA zone driver, FCM push dispatcher.
Data Integrity & Caching	PostgreSQL 15, SQLAlchemy, Alembic, Redis	ACID relational database store with 3NF schema, immutable audit logging, Redis session cache, and WebSocket pub/sub broker.

2. Entity-Relationship (ER) Diagram & Schema

The EchoSphere v2.2 relational database schema is structured in **Third Normal Form (3NF)** to eliminate redundancy, enforce strict referential integrity, and maintain a complete audit trace of all communication activities.



Comprehensive Data Dictionary & Entity Specifications

Entity: User — *Stores user profiles, credentials, role assignments, and department affiliations.*

Column	Data Type	Constraints	Description
id	INTEGER	PK, Auto	Unique user identifier
name	VARCHAR(100)	NOT NULL	Full name of the user
email	VARCHAR(150)	UQ, NOT NULL	Official campus email address
hashed_password	VARCHAR(255)	NOT NULL	Bcrypt hashed security password
role_id	INTEGER	FK -> Role.id	Assigned RBAC system role
department_id	INTEGER	FK -> Dept.id	Associated department reference
is_active	BOOLEAN	DEFAULT TRUE	Account active/disabled flag

Entity: Announcement — *Core entity representing campus communications across approval states.*

Column	Data Type	Constraints	Description
id	INTEGER	PK, Auto	Unique announcement identifier
title	VARCHAR(200)	NOT NULL	Announcement headline/subject
content	TEXT	NOT NULL	Full announcement body text
author_id	INTEGER	FK -> User.id	Creator / author user reference
category_id	INTEGER	FK -> Cat.id	Category taxonomy classification
department_id	INTEGER	FK -> Dept.id	Target department filter (NULL = All)
priority	VARCHAR(20)	DEFAULT 'NORMAL'	Urgency priority level (EMERGENCY, HIGH, NORMAL)
status	VARCHAR(30)	DEFAULT 'DRAFT'	Lifecycle status (DRAFT, PENDING, APPROVED, REJECTED, ARCHIVED)

Entity: SpeakerQueue — *Manages audio synthesis jobs and physical PA speaker play queues.*

Column	Data Type	Constraints	Description
id	INTEGER	PK, Auto	Queue item identifier
announcement_id	INTEGER	FK -> Ann.id	Target announcement for audio playback
audio_url	VARCHAR(255)	NOT NULL	Path to synthesized MP3/WAV file
priority	INTEGER	DEFAULT 0	Play queue priority weight
status	VARCHAR(20)	DEFAULT 'PENDING'	Execution state (PENDING, PLAYING, COMPLETED, FAILED)

Entity: AuditLog — *Immutable security log tracking system actions, approvals, and logins.*

Column	Data Type	Constraints	Description
id	INTEGER	PK, Auto	Audit entry identifier
user_id	INTEGER	FK -> User.id	Actor performing the action
action	VARCHAR(100)	NOT NULL	Action keyword (LOGIN, CREATE, APPROVE, ARCHIVE)
resource_type	VARCHAR(50)	NOT NULL	Target entity table name
ip_address	VARCHAR(45)	NOT NULL	Client IP address for security tracing

3. System Methodology & Operational Workflows

EchoSphere v2.2 follows an **Agile Software Engineering Methodology** combined with strict **Role-Based Access Control (RBAC)**, a unified **DevAdmin UI Synchronization Rule**, and an automated **AI-Augmented Announcement Lifecycle Pipeline**.

3.1 Role-Based Access Control (RBAC) & Feature Matrix

EchoSphere enforces a strict authorization hierarchy across 6 distinct user roles to guarantee security, departmental privacy, and controlled broadcast authority:

Feature / Capability	DevAdmin	Principal	College Admin	HOD	Teacher	Student
Create Announcement	Yes	Yes	Yes	Yes	Yes	No
AI Voice/STT Drafting	Yes	Yes	Yes	Yes	Yes	No
Approve College Notices	Yes	Yes	Yes	No	No	No
Approve Dept Notices	Yes	Yes	Yes	Yes	No	No
Emergency Override	Yes	Yes	No	No	No	No
Speaker Queue Control	Yes	Yes	Yes	View Only	No	No
View Broadcast Feed	Yes	Yes	Yes	Yes	Yes	Yes
System Audit Logs	Yes	View Only	No	No	No	No

UI CONSISTENCY & ROLE PROPAGATION GUARANTEE

MANDATORY WORKSPACE RULE: Any visual UI/UX design enhancement or layout component improvement implemented on the Developer Administrator (devadmin) interface must automatically be propagated across all other role pages (Principal, College Admin, HOD, Teacher, Student). Permission boundaries and role restrictions remain strictly enforced.

3.2 Announcement Lifecycle & Audio Dispatch Methodology

The end-to-end communication lifecycle inside EchoSphere v2.2 operates through a multi-stage sequential state machine:

- **1. Draft Creation & AI Composition:** Author drafts an announcement manually or speaks via Whisper Speech-to-Text. The AI Text Expander automatically converts rough notes into rich, structured content.
- **2. AI Content Moderation & Emergency Triage:** The AI Microservice screens the text for spam/toxicity and evaluates emergency markers. High urgency triggers an emergency priority escalation.
- **3. Multi-Level Approval Routing:** Depending on scope, the announcement is routed to HOD (Department level) or Principal/College Admin (Campus level) for review and approval.
- **4. Audio Synthesis & Speaker Queueing:** Upon approval, the system generates multi-lingual neural TTS MP3 audio and pushes a job to the Speaker Queue with appropriate priority weights.
- **5. Multi-Channel Synchronization & Dispatch:** Simultaneously dispatches FCM mobile push notifications, updates WebSocket live feeds, and triggers physical/virtual PA speaker audio output.
- **6. Archival & Audit Trailing:** Expired announcements are automatically archived with full audit logging recorded in PostgreSQL for compliance and safety analysis.

LAYOUT RESPONSIVENESS & ZERO OVERFLOW GUARANTEE

ZERO OVERFLOW CONSTRAINT: All mobile and desktop screens are engineered using fluid layout widgets (Expanded, Flexible, SingleChildScrollView, Wrap) to guarantee zero layout overflow errors across all screen sizes (320px compact phones to 4K desktop displays).